

- 9 Fig. 9 shows a sketch of the region OPQ of the Argand diagram defined by

$$\{z : |z| \leq 4\sqrt{2}\} \cap \{z : \frac{1}{4}\pi \leq \arg z \leq \frac{1}{3}\pi\}.$$

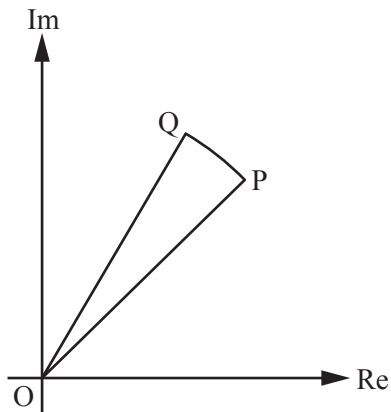


Fig. 9

- (i) Find, in modulus-argument form, the complex number represented by the point P. [2]
- (ii) Find, in the form $a + ib$, where a and b are exact real numbers, the complex number represented by the point Q. [3]
- (iii) **In this question you must show detailed reasoning.**

Determine whether the points representing the complex numbers

- $3 + 5i$
- $5.5(\cos 0.8 + i \sin 0.8)$

lie within this region.

[4]